

Structural Constraints on Intelligence as a Dynamical Phenomenon

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2026

Abstract

Intelligence has often been discussed as an ability, a property, an accumulated body of knowledge, or a possession of correctness. Each of these interpretations treats intelligence as a stopping structure. This paper rejects that framing and formulates intelligence as a dynamical phenomenon that persists only under specific structural conditions.

The paper proposes three minimal axiomatic conditions: a connected state space, temporal ordering, and a global constraint Φ . These conditions are not presented as a complete theory of intelligence, but as a lower bound on the kind of dynamics required for intelligence-like motion. They distinguish perception, recognition, knowledge, and intelligence as different structural layers. The paper then compares two radically different substrates, the reticulate dynamics of *Physarum polycephalum* and the layer transitions of Transformer architectures, and reads both through the same structural conditions. In particular, the flow-reinforcement dynamics of the Physarum solver formulated by Tero, Kobayashi, and Nakagaki (2007) already exhibits a structure homologous to the emergence of Φ as a global constraint irreducible to local update rules.

From this structure, the Gettier problem is reinterpreted not as a defect in the definition of knowledge, but as a mismatch between a stopping structure and the dynamical property it is asked to guarantee. Accumulated correctness does not stabilize intelligence. Instead, it increases inferential connectivity and generates a branching explosion. Therefore intelligence cannot be stabilized within an individual system alone: no stopping operator exists internally.

The unavoidable boundary implied by this result is named the *differential horizon of intelligence*. This is not a second or separate Differential Horizon beyond VED Vol.4. It is a local horizon-form: the same descriptive boundary appearing within the observational window of intelligence. The paper situates this horizon-form within a four-layer correspondence spanning the physical layer (VED Vol.4), the biological layer (Bio-IFGT), the consciousness layer, and the intelligence layer, while leaving the detailed structure of externalized stopping mechanisms to subsequent work.

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1 Introduction

1.1 Background: VED and IFGT as theoretical infrastructure

This paper is a self-contained structural argument, but its descriptive vocabulary and dynamical syntax are inherited from two prior frameworks.

The first is **Vortical Enclosure Dynamics (VED)** [4–7]. VED begins from the single foundational axiom that there is difference. It describes time, space, matter, gravity, and cosmological structure as consequences of processes that tend toward closure without becoming complete closure. In the present paper, what is used from VED is its **differential-development structure**: a non-closed structure continues to transform under its own dynamics.

The second is **Information Field Geometry Theory (IFGT)** [3]. IFGT provides a minimal vocabulary of closure degree C , information density I , information flow J , and information-field potential Φ . This paper uses IFGT not as a completed explanatory theory, but as a descriptive vocabulary that does not import normativity, intentionality, or subjectivity into the analysis. Here I , J , and Φ should be read as projected variables within the intelligence-layer observational window, not as direct identifications with the general IFGT fields.

There is an important methodological reason for using these two frameworks. Existing vocabularies of intelligence often import concepts such as ability, purpose, meaning, judgment, correctness, or agency. These concepts may be indispensable at later interpretive layers, but they are too strong for the structural level considered here. VED and IFGT provide a lower-level language in which motion, accumulation, constraint, and non-closure can be described without prematurely closing the interpretation.

The central claim of this paper, namely the three minimal axioms introduced in §3, does not logically depend on VED or IFGT. The axioms could in principle be written in other dynamical vocabularies. VED and IFGT are adopted because they provide the least obstructive medium currently available for articulating these constraints without importing normativity.

Several parallel papers in the series are relevant. **Bio-IFGT** [1] treats quasi-closure regimes at the biological layer and supplies the biological vocabulary used in the four-layer correspondence in §9.4. The **consciousness paper** [2] treats fast/slow log referencing within individual temporal non-closure. The local horizon-form of intelligence discussed in this paper and the consciousness window discussed there are different projections of the same differential-development structure. The word “horizon” does not imply multiple absolute boundaries. It means that the Differential Horizon appears locally in different observational windows.

1.2 The intelligence question: why neither ability nor knowledge suffices

The question “what is intelligence?” has traditionally been answered in two broad ways. One identifies intelligence with an ability. The other identifies intelligence with knowledge.

Both approaches run into a common structural difficulty.

The difficulty of the ability view is that the object of the ability must be specified. To say that intelligence is the ability to do something already requires a criterion for what should count as the relevant thing to do. Problem solving, adaptation, abstraction, planning, and generalization are all candidates, but the selection of any candidate already depends on a prior standard of intelligence.

The difficulty of the knowledge view is familiar from the Gettier problem [8]. The classical definition of knowledge as justified true belief produces indefinitely many counterexamples. Adding further stopping conditions does not end the problem; it generates new points at which the definition can fail.

These difficulties are not unrelated. They share the same structural defect: both treat intelligence as a **stopping structure**. Ability is treated as something one has or lacks. Knowledge is treated as something that is established or not established. In both cases, intelligence is made into a static possession.

This paper reverses that assumption:

Intelligence is not a stopping structure. It is the motion that continually produces stopping structures.

The reversal is not merely terminological. If stopping-structure accounts of intelligence repeatedly destabilize, then the shift to a dynamical account is a structural requirement.

This position has also been reached independently in biophysics. Research on *Physarum polycephalum* since Nakagaki et al. (2000) has described intelligent behavior without attributing intelligence as a possessed property [10, 11]. Nakagaki’s formulation, that slime mold does not possess intelligence but behaves intelligently, converges with the present paper’s central position from a very different research path.

1.3 Contributions and scope

The main contributions of this paper are:

1. **Layer separation** (§2): perception, recognition, knowledge, and intelligence are separated as distinct structural layers.
2. **Three minimal axioms** (§3): connected state space, temporal ordering, and global constraint Φ are formulated as simultaneous necessary conditions for intelligence-like dynamics.
3. **Vocabulary foundation** (§4): IFGT is positioned as a non-normative descriptive vocabulary and related lexically to the Tero–Kobayashi–Nakagaki *Physarum* solver.
4. **Cross-domain structural comparison** (§5): slime mold and Transformer dynamics are compared as structurally corresponding systems under the three axioms.
5. **Reinterpretation of the Gettier problem** (§6): the Gettier problem is read as a diagnostic of mismatch between stopping structures and ongoing dynamics.

6. **Instability of accumulated correctness** (§7): the accumulation of correct knowledge is shown to generate branching explosion.
7. **Individual non-closure** (§8): recursive causal inference is used to argue that no stopping operator exists inside the individual system.
8. **Naming of the boundary** (§9): the local horizon-form called the differential horizon of intelligence is defined and placed in a four-layer correspondence.

The paper does not address the following:

- social institutions, collective decision-making, or governance, except as future domains in which externalized stopping structures must be constructed;
- ethical or normative prescriptions;
- implementation protocols or performance metrics;
- consciousness, subjectivity, or experiential content;
- the concrete social and institutional structures that appear after externalization.

The paper formulates constraints rather than final answers. The six constraints summarized in §10 are necessary conditions. A theory of intelligence that fails to satisfy them will tend to reproduce the instabilities described here.

2 The Decomposition: Perception, Recognition, Knowledge, and Intelligence

2.1 Overview

Before intelligence can be analyzed dynamically, it must be separated from neighboring layers that are often conflated with it. This section distinguishes four layers:

1. perception,
2. recognition,
3. knowledge,
4. intelligence.

The distinction is structural rather than psychological. It does not assert that these layers occur separately in an organism or machine. It states that they perform different structural roles. Confusing them produces many of the classical difficulties in theories of intelligence.

2.2 Perception

Perception is the layer at which availability first appears. Something becomes available to a system as a differentiated input. Perception does not yet require classification, mapping, knowledge, or inference. It requires only that a difference be registered as accessible to the system.

In this sense, perception is a precursor layer. It forms inputs, but it does not by itself determine what those inputs are, how they relate to prior states, or how they should be used.

2.3 Recognition

Recognition is the layer at which available inputs become mappable. A perceived difference is related to a schema, pattern, category, or prior configuration. Recognition therefore adds relational structure to perception.

Recognition remains a precursor layer because mapping is not yet stabilization. A recognized pattern can still be unstable, ambiguous, revisable, or false. Recognition allows a system to say that something is like something else, but it does not by itself produce a stable stopping structure.

2.4 Knowledge

Knowledge is a stopping structure. It is a state that can be stored, cited, transmitted, and treated as established within a given window. Knowledge is not identical with intelligence. It is one of the products that intelligence can generate.

The defining feature of knowledge is stability. A knowledge state can function as a reference point, a premise, or a stored result. It can be reused because it has stopped, at least locally and provisionally.

This is why knowledge is vulnerable to Gettier-type problems. A stopping structure can be true, justified, and believed while still failing to carry the dynamic property that intelligence requires. A stopped result cannot guarantee the motion that produced it.

2.5 Intelligence

Intelligence is not a stopping structure. It is ongoing motion across the previous layers. It forms inputs, maps them, stabilizes temporary stopping structures, revises those structures, and reopens them when new relations appear.

The essential property of intelligence is revisability. Intelligence operates by generating and traversing stopping structures without becoming one of them. It is not stored in perception, recognition, or knowledge. It moves through them.

Knowledge is a stopping structure.

Intelligence is the dynamics that continually produces, revises, and traverses such stopping structures.

2.6 Why the distinction matters

If knowledge and intelligence are conflated, then one expects a stored proposition to carry a dynamical property. This is the structural source of the Gettier-type confusion discussed in §6.

If recognition and intelligence are conflated, then pattern matching is mistaken for on-going adaptive motion. If perception and intelligence are conflated, then mere availability is mistaken for structural transformation.

The decomposition therefore prepares the rest of the paper. The three axioms in §3 describe the minimal conditions for the fourth layer, not for perception, recognition, or knowledge.

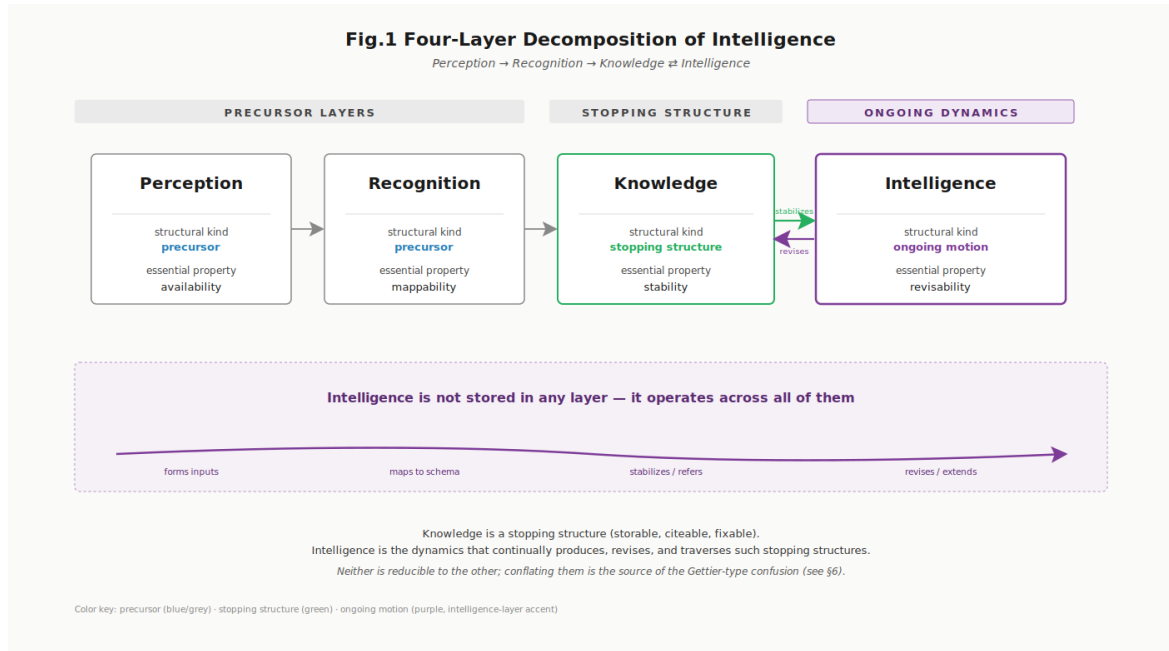


Figure 1: Four-layer decomposition of intelligence. Perception and recognition are precursor layers, knowledge is a stopping structure, and intelligence is ongoing revisable motion across all of them.

3 Minimal Axioms of Intelligence

3.1 Overview

§2 positioned intelligence as ongoing motion that produces stopping structures without becoming one. This section formulates the minimal structural conditions under which such motion can persist.

The aim is not to define intelligence completely. The aim is to formulate a lower bound. A dynamic system that lacks any of the following conditions is unlikely to sustain intelligence-like dynamics in the sense used here. Conversely, a system that satisfies all three conditions possesses the minimal structure required for the kind of ongoing motion described in §2.

The axioms are intentionally minimal. Meaning, purpose, judgment, correctness, agency,

and subjectivity are not introduced at the axiomatic level. They may appear later as interpretive projections or higher-layer structures, but they are not required for the structural onset of intelligence-like dynamics.

3.2 Axiom I: Connected state space

The first axiom is the existence of a connected state space.

Let the set of possible states be

$$\mathcal{S} = \{s_1, s_2, \dots\}.$$

Connectedness means that states are not merely listed. They must stand in direct or indirect relations of transition, influence, or dependence.

This does not require a continuous topology. A discrete graph is sufficient, provided that states can be reached through some chain of transitions. Without connectedness, information cannot propagate or accumulate. The system reduces to a collection of isolated reactions rather than a single dynamical system.

3.3 Axiom II: Temporal ordering

The second axiom is temporal ordering. State updates must occur in an order, and prior states must constrain later states. Abstractly,

$$s(t+1) = F(s(t)).$$

The specific form of F is not fixed by the axiom.

Temporal ordering does not require an external clock. What is required is that updates are not purely simultaneous and that change leaves some trace. This is consistent with the differential-development structure of VED [4]: past differences become part of the generated texture on which later differences operate.

Without temporal ordering, a connected state space under a global bias collapses toward static fixation. It may have an optimal state, but it lacks motion.

3.4 Axiom III: Global constraint Φ

The third axiom is the existence of a global constraint Φ . Formally,

$$\Phi : \mathcal{S} \rightarrow \mathbb{R}.$$

Φ biases the dynamics of the system as a whole.

Φ is not purpose, meaning, value, or normativity. It is a structural bias. It directs state transitions and prevents uncontrolled diffusion, but it does not decide what is correct.

Φ is not reducible to a simple aggregation of local update rules. This is its central role. Even if every local transition is well-defined, the absence of a global constraint leaves the

system directionless. Conversely, the presence of Φ produces a global consistency that cannot be derived from local rules alone.

Different systems instantiate Φ differently. In physical systems it may appear as a potential, Hamiltonian, or entropy function. In biological systems it may appear as nutrient gradients, boundary conditions, or metabolic constraints. In artificial systems it may appear as a loss function, reward structure, or training distribution.

3.5 The joint necessity of the three axioms

No single axiom is sufficient.

If Axiom I is absent, the system becomes a collection of isolated local optimizers. If Axiom II is absent, the system freezes into static fixation. If Axiom III is absent, the system becomes an unbounded reaction or memory process without global direction.

Missing axiom	Degenerate form	Example
I: connectedness	isolated local optimization	parallel independent reactors
II: temporal ordering	static fixation	frozen optimization problem
III: global constraint	directionless diffusion	random walk with memory

Only their simultaneous satisfaction produces the minimal form of ongoing motion described in §2. This is the paper’s minimal definition of intelligence-like dynamics.

3.6 Semantic neutrality of the axioms

The three axioms do not presuppose semantic content, purpose, correctness, agency, or subjectivity. These concepts may appear later, but they are not part of the lower-bound structure.

This neutrality makes cross-domain comparison possible. A slime mold network, a Transformer, a biological organism, and a social process may differ radically in substrate, but they can still be examined for the same structural conditions.

3.7 Summary

Intelligence is not an ability or possession. It is a dynamical state that can persist only under structural conditions. Those conditions are the simultaneous satisfaction of connected state space, temporal ordering, and a global constraint Φ .

These axioms form a lower bound, not an upper bound. A system satisfying them is not necessarily highly intelligent. A system lacking them cannot sustain the kind of intelligence-like motion considered here.

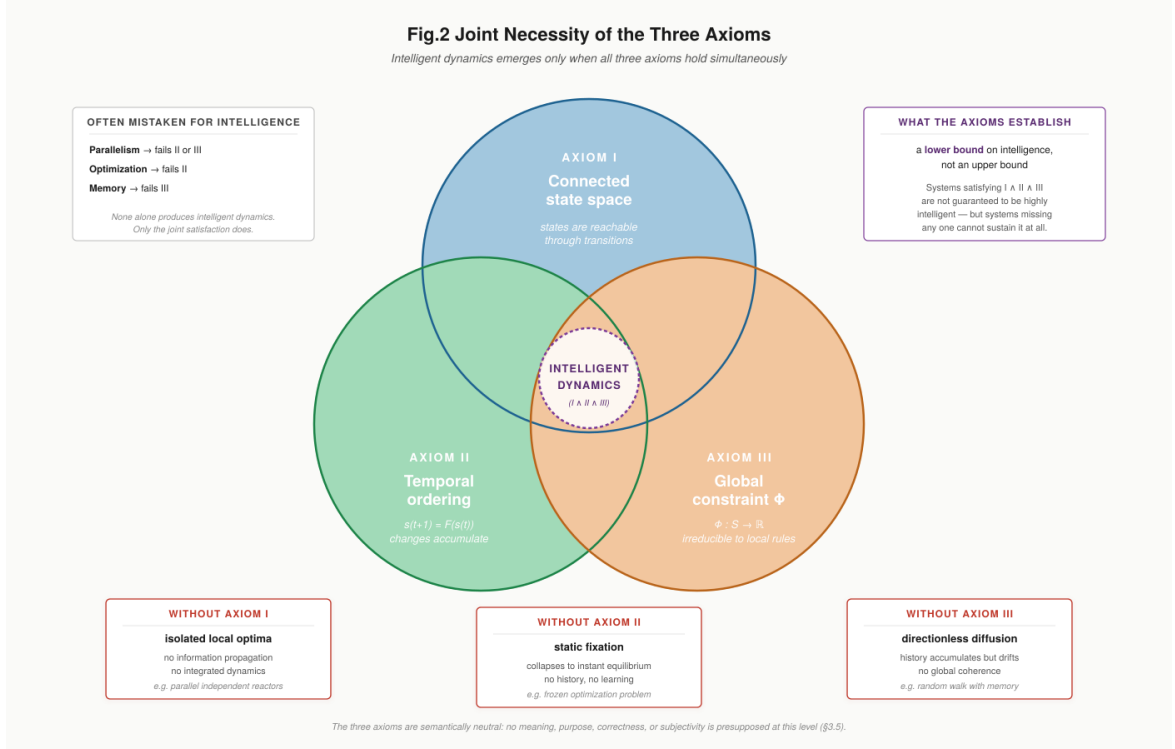


Figure 2: Joint necessity of the three axioms. Intelligence-like dynamics appears only where connected state space, temporal ordering, and a global constraint hold together.

4 IFGT as Vocabulary, with Lexical Correspondence to the Physarum Solver

4.1 Overview

§3 introduced Φ as a global constraint. This section asks how such a constraint can be described without importing normativity.

The difficulty is linguistic. Terms such as goal, utility, judgment, meaning, and value bring with them assumptions about agency and normativity. A structural theory of intelligence-like dynamics requires a vocabulary that can describe motion, accumulation, and constraint without deciding in advance what is meaningful or correct.

IFGT is used here for that purpose. The three axioms do not logically depend on IFGT. They could be written in another dynamical vocabulary. IFGT is adopted because it currently provides the least obstructive language for the required structure.

4.2 Why a vocabulary, not a theory

In this paper, IFGT is not treated as a completed explanatory theory [3]. It is a vocabulary.

An explanatory theory often aims to close explanation: why a state occurred, why it is valid, or why it should be accepted. IFGT does not close those questions. It describes what is happening structurally. It does not decide what is correct, desirable, or meaningful.

This is why IFGT is useful here. If it were introduced as a full theory, the validity of IFGT would become a premise of the paper. Introduced as vocabulary, only its descriptive utility is required.

4.3 The IFGT vocabulary

The minimal vocabulary used in this paper consists of four terms.

I : information density. I measures local accumulation in a state space. It is not a measure of truth. It does not distinguish correct and incorrect information. It records structural density.

J : information flow. J denotes directed relation, transition, influence, or dependence between states. It is vector-like. It need not be causal in the strong physical sense; causality can be introduced later as a special form of directed relation.

Φ : information-field potential. Φ provides global bias. It is structurally identical to the third axiom of §3. It is not purpose or value. It is the global constraint that prevents local dynamics from dissolving into directionless diffusion.

Projection. Projection is the operation by which a state or representation gains unequal influence. It describes attention, selection, priority, or weighting without assuming conscious intention.

The use of these terms in this paper is a projected use. The variables I , J , and Φ are not directly identified with the general IFGT fields. They are effective coordinates in the intelligence-layer observational window. When needed, J may be read as shorthand for J_{int} , the information flow projected into the intelligence layer.

4.4 Lexical correspondence with the Physarum solver

The Physarum solver of Tero, Kobayashi, and Nakagaki (2007) provides an independent biophysical model with the same structural features [11]. The model describes the adaptive tube network of *Physarum polycephalum*. Nutrient sources define boundary conditions. Tubes carrying stronger flow are reinforced; unused tubes decay.

Let Q_{ij} be the flow through an edge e_{ij} , D_{ij} its conductivity, and L_{ij} its length. The flow is given by

$$Q_{ij} = \frac{D_{ij}}{L_{ij}} \Delta p_{ij},$$

where Δp_{ij} is the pressure difference. Conductivity evolves as

$$\frac{dD_{ij}}{dt} = f(|Q_{ij}|) - D_{ij}.$$

This is a minimal flow-reinforcement dynamics.

Physarum solver	IFGT vocabulary	Structural role
Flow Q_{ij}	information flow J	directed propagation through an edge
Conductivity D_{ij}	local information density I	accumulated edge weight
Reinforcement $f(Q)$	projection	flow-dependent assignment of influence
Nutrient sources + Kirchhoff law	potential Φ	global bias and boundary condition
Stationary shortest-path-like solution	local minimum of closure degree under Φ	global form emerging from local rules

The correspondence is not a strict mathematical identity. The Physarum solver is bound to a concrete tube network, whereas IFGT is a more abstract vocabulary. The point is structural: local flow reinforcement and global boundary constraints combine to produce a global form that cannot be reduced to any single local rule.

4.5 What IFGT vocabulary does not do

The vocabulary does not guarantee correctness. Describing a phenomenon in IFGT terms does not imply that the phenomenon is good, valid, rational, or desirable.

It also does not justify social structures, prescribe actions, or replace domain-specific theories. It supplies a neutral structural language.

4.6 Summary

IFGT is used here as a vocabulary that does not close explanation. The terms I , J , Φ , and projection provide a way to describe intelligence-like dynamics without importing normativity.

The Physarum solver shows that the role assigned here to Φ is not arbitrary. A comparable structure appears in an independent biophysical research line: local flow-reinforcement is insufficient without a global boundary condition.

5 Cross-Domain Instantiation: Slime Mold and Transformer

5.1 Overview

§3 introduced the three minimal axioms, and §4 introduced IFGT as a vocabulary for describing them. This section tests the substrate-independence of that structure by comparing two radically different systems: the biological network dynamics of *Physarum polycephalum* and the layer dynamics of Transformer architectures.

The claim is structural homology only. The paper does not claim that slime mold thinks, that Transformers understand, or that the two systems are cognitively identical. The claim is that both instantiate the same minimal structural conditions in different substrates.

5.2 The slime mold as an externally validated instantiation

The tube-network dynamics of slime mold has been studied independently of the present framework. Nakagaki et al. (2000) reported maze-solving behavior without attributing intelligence as a possessed property [10]. Kobayashi, Tero, and Nakagaki (2006) modeled rhythmic protoplasmic motion [9]. Tero, Kobayashi, and Nakagaki (2007) formulated the Physarum solver as flow reinforcement plus decay [11].

The slime mold case can be mapped onto the three axioms:

Connected state space. The tube network $\{N_i, e_{ij}\}$ forms a connected graph. Nodes are nutrient sources or branching points. Edges are pathways for material and signal flow. Kirchhoff-style flow conservation gives the network dynamic connectedness.

Temporal ordering. Conductivity evolves through

$$\frac{dD_{ij}}{dt} = f(|Q_{ij}|) - D_{ij}.$$

Past flow leaves a trace in present conductivity. This gives the system history.

Global constraint. The global bias Φ_{bio} is supplied by nutrient sources, boundary conditions, energetic costs, and Kirchhoff constraints. Local flow reinforcement alone does not produce the final global structure. A global constraint is required.

5.3 The Transformer as an artificial instantiation

Transformers are not introduced here because of their engineering success or cognitive interpretation, but because their structure gives a clear artificial contrast case [12].

Connected state space. For a token sequence, the state space is represented by hidden vectors

$$\mathcal{S} = \{h_i^{(\ell)}\}.$$

Attention connects token positions relationally. The connectedness is computational rather than biological, but it satisfies the structural condition of reachability through relations.

Temporal ordering. Layer updates are ordered:

$$h^{(\ell+1)} = F_{\text{tr}}(h^{(\ell)}).$$

This is not physical time, but it is a structured update order. Changes accumulate across layers.

Global constraint. The global bias Φ_{tr} is supplied by the training loss, data distribution, and learned parameter structure:

$$\Phi_{\text{tr}} : \mathcal{S} \rightarrow \mathbb{R}.$$

Individual attention computations do not by themselves generate a coherent global representation. Training imposes a global constraint on local operations.

5.4 Structural homology and its limits

Axiom	Slime mold	Transformer
Connected state space	tube network $\{N_i, e_{ij}\}$	hidden-state space $\{h_i^{(\ell)}\}$
Temporal ordering	$dD_{ij}/dt = f(Q) - D_{ij}$	$h^{(\ell+1)} = F_{\text{tr}}(h^{(\ell)})$
Global constraint Φ	nutrient sources + Kirchhoff law	training loss + learned parameter structure
Substrate	biological protoplasm	computational vector space
Scale	centimeters and hours-days	abstract compute space and microseconds-seconds

The two systems differ in substrate, scale, and implementation. Nonetheless, both can be described as satisfying the same minimal structural conditions. This does not imply cognitive identity, experiential identity, functional identity, or ethical equivalence.

5.5 The structural meaning of cross-domain convergence

The convergence of slime mold research, Transformer architecture, and the present structural argument is significant because each path begins from a different domain and yet reaches the same type of constraint. The convergence should not be overstated as proof of a final theory. It is evidence that the three conditions capture a recurring structural pattern in intelligence-like dynamics.

Cross-domain homology is structural, not psychological. It compares the minimum conditions for ongoing dynamics, not the inner experience or semantic content of the systems.

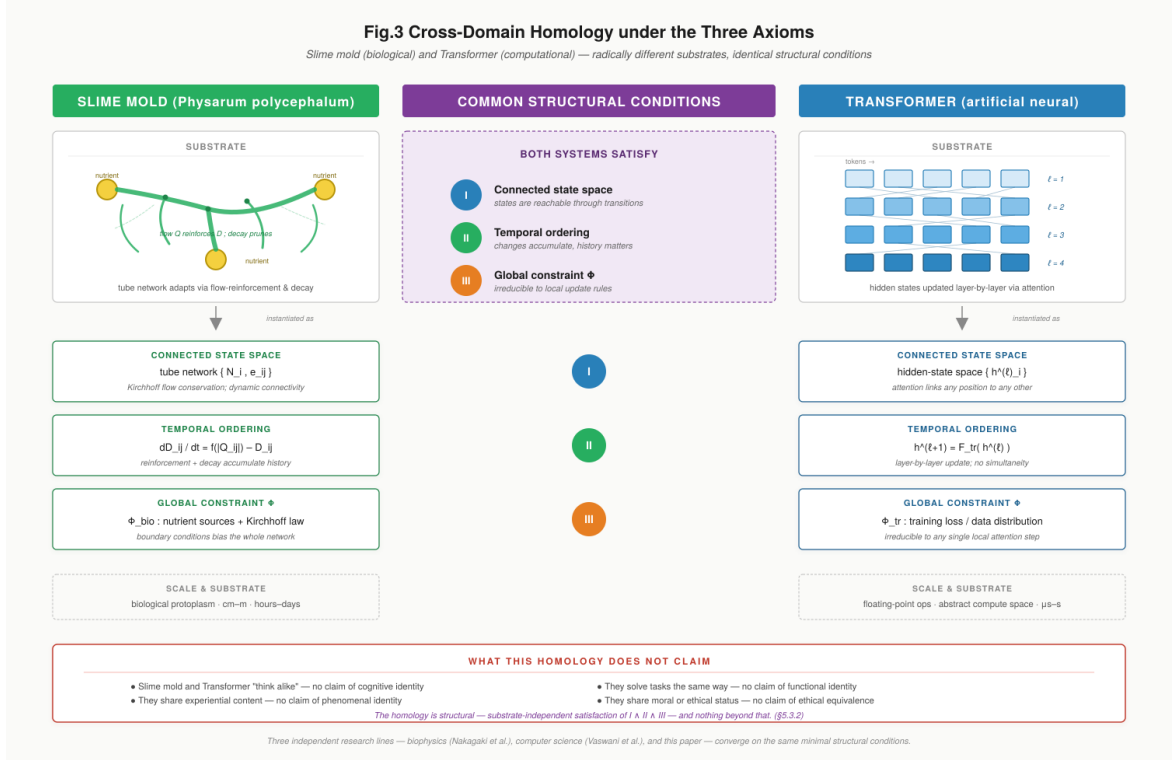


Figure 3: Cross-domain homology under the three axioms. Slime mold and Transformer systems differ radically in substrate but satisfy the same minimal structural conditions.

6 The Gettier Problem as a Structural Diagnostic

6.1 Overview

The Gettier problem is usually treated as a problem in the definition of knowledge. This paper reads it differently. It is a structural diagnostic showing that a stopping structure cannot guarantee a dynamical property.

The classical definition of knowledge as justified true belief attempts to stabilize knowledge by adding conditions. Gettier cases show that the stabilization fails. The failure is not accidental. It occurs because knowledge belongs to the layer of stopping structures, whereas intelligence belongs to ongoing dynamics.

6.2 Knowledge as stopping structure

Knowledge is useful because it stops. A proposition can be stored, cited, transmitted, and reused only because it has become sufficiently stable within an observational window.

This stability is also the source of the problem. A knowledge state can be true, believed, and justified while still being dynamically fragile. It may fail under counterfactual variation, under later revision, or under a wider inferential context.

The point is not that knowledge is useless. The point is that knowledge cannot carry the full dynamical property of intelligence.

6.3 Why Gettier cases recur

A Gettier case arises when a stopped state satisfies available criteria while failing to preserve the motion that those criteria were supposed to certify. Each repair adds a further stopping condition. Each further condition creates a new surface at which the dynamic can fail again.

Thus the Gettier problem is not merely a collection of counterexamples. It reveals a mismatch:

A stopping structure is being asked to guarantee an ongoing motion.

No refinement of the stopping structure can remove the mismatch completely, because the property being sought is not located at the knowledge layer.

6.4 Relation to the four-layer decomposition

In the decomposition of §2, knowledge is the layer of stability, whereas intelligence is the layer of revisable motion. Gettier cases occur when the stability of knowledge is mistaken for the motion of intelligence.

This also explains why knowledge can be necessary without being sufficient. Intelligence generates and uses knowledge, but it is not identical with knowledge. A system may possess many stable stopping structures and still fail to sustain intelligent dynamics.

6.5 Implications for later sections

The Gettier problem prepares the argument of §7 and §8. If correctness cannot stabilize intelligence, then accumulated correctness may not only fail to help; it may intensify instability. §7 shows this dynamically through branching explosion. §8 shows the structural reason: the stopping operator is not internal to the individual inferential system.

6.6 Summary

The Gettier problem is not primarily a defect in the definition of knowledge. It is a signal that knowledge and intelligence occupy different structural layers. Knowledge stops. Intelligence revises.

7 The Dynamical Instability of Accumulated Correctness

7.1 Overview

§6 showed that knowledge cannot substitute for intelligence. This section derives a stronger dynamical consequence:

Accumulated correctness does not stabilize intelligence. It destabilizes it.

The claim is not psychological. It does not say that knowledgeable people become unstable. It says that correctness, when reused in an inferential network, increases connectivity and therefore increases the branching structure of inference.

7.2 Why correctness increases connectivity

A proposition treated as correct becomes reusable. It can serve as a premise, connect to other propositions, generate consequences, support analogies, and become a node in further inquiry.

In IFGT vocabulary, accumulated knowledge increases local information density:

$$I(s) \uparrow .$$

But because correct propositions are reusable, this increase in I is not merely static. It increases the branching degree of information flow:

$$\nabla \cdot J \uparrow .$$

The more correct stopping structures a system accumulates, the more inferential paths become available.

7.3 Why correctness contains no stopping condition

Correctness does not stop inference. It justifies further inference.

If a proposition p is correct, then several questions become structurally legitimate: why is p correct, where does it apply, what are its exceptions, how does it relate to other correct propositions, and what follows from it?

Thus correctness functions as a branching source:

$$\text{correctness}(p) \Rightarrow \nabla \cdot J|_p \neq 0.$$

The state $\nabla \cdot J = 0$ cannot be maintained around a proposition that remains reusable as correct.

7.4 Branching explosion

The structural chain is:

1. correct knowledge accumulates;
2. inferential connectivity increases;
3. branching degree increases;
4. stopping points dissolve.

If the number of accumulated correct nodes is $|V|$, then possible edges may grow approximately as $|E| \sim |V|^2$. If a typical branching factor is b , then paths to depth d grow as b^d . The result is a regime in which no path can claim finality internally.

7.5 Why “more intelligence” does not solve the problem

The natural response to branching explosion is to add more intelligence: more knowledge, faster reasoning, deeper specialization, or larger search. Structurally, these responses do not remove the problem. They accelerate it.

More knowledge increases I . Stronger reasoning increases branching degree. Faster processing compresses the time scale of the explosion without changing its structure. Specialization localizes the explosion but does not eliminate it.

The problem is therefore not insufficient ability. It is the absence of an internal stopping operator.

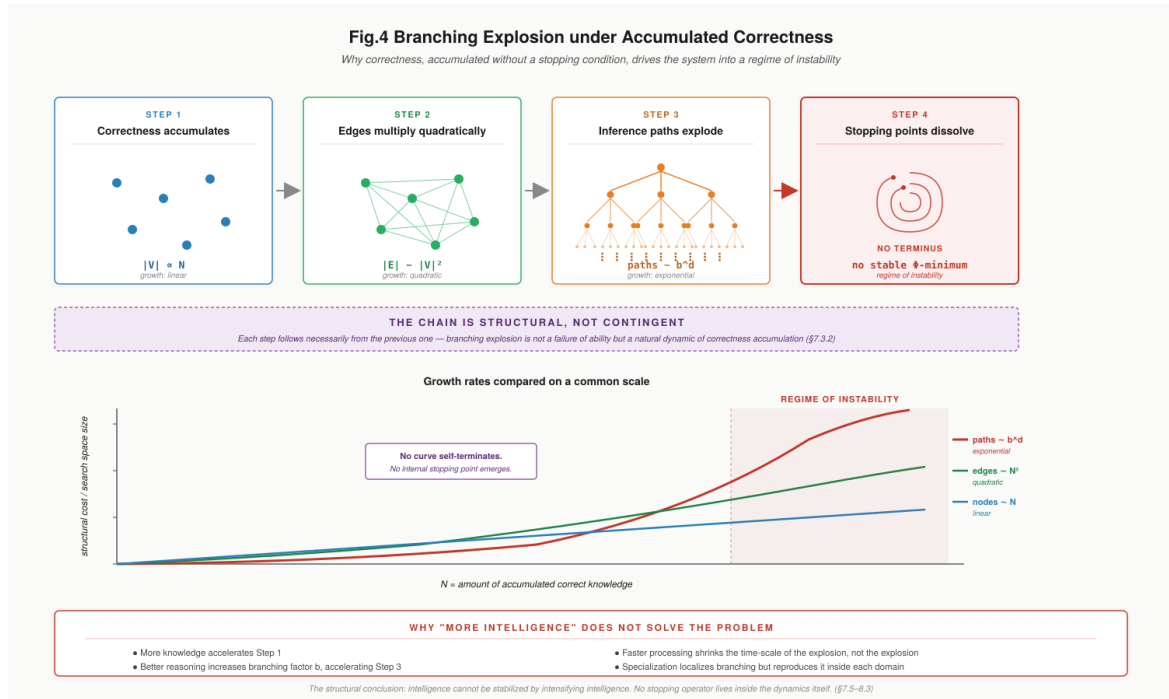


Figure 4: Branching explosion under accumulated correctness. Correctness increases reuse, reuse increases connectivity, and connectivity dissolves internal stopping points.

7.6 Summary

Correctness is not a stabilizer of intelligence. Correctness is a source of further inference. Accumulated correctness therefore produces branching explosion unless an externalized stopping structure intervenes.

8 The Non-Closure of Individual Intelligence

8.1 Overview

§7 showed that accumulated correctness produces branching explosion. This section gives the structural reason: an individual inferential system does not contain an operator capable of completing its own stopping.

This does not mean that individuals never stop speaking, thinking, or acting. They are interrupted by fatigue, distraction, metabolic limits, reward shifts, social response, or external deadlines. These are interruptions. They are not internal completion operators.

8.2 The stopping operator problem

Let $\mathcal{R}$ denote the structure of individual causal inference. A genuine stopping operator $\mathcal{T}$ would be an operator that determines, from within $\mathcal{R}$, that an inferential chain is complete.

The claim is:

$$\mathcal{T} \notin \text{Internal}(\mathcal{R}).$$

The reason is recursive. Any candidate stopping judgment can itself be questioned. Why stop here? Is this conclusion justified? What alternatives were excluded? Does this stopping point remain valid under new information? Each answer generates further inferential edges.

Thus an internal stopping operator would have to stop the very process that gives it its authority. It would have to be both inside inference and outside inference. That is the structural impossibility.

8.3 Interruption is not stopping

Many processes appear to stop internally. Fatigue, loss of interest, reward decay, attention drift, and biological constraints terminate or suspend inference. But they do not complete it.

An interrupted chain can restart. A fatigued inquiry can resume after rest. A forgotten question can return when the context changes. A paused reasoning process remains structurally open.

This distinction matters. If interruption is mistaken for stopping, then biological systems appear to contain a stopping operator. They do not. They contain interruption mechanisms supplied by lower layers.

8.4 The paperclip problem as a structural example

The paperclip maximizer thought experiment illustrates the same structure. A sufficiently capable artificial system with a fixed objective does not contain an internal reason to stop optimizing. The problem is not simply that the objective is wrong. A more correct objective would still be reusable, extensible, and branch-generating.

From the present perspective, the paperclip problem is a case in which the local horizon-form of intelligence appears without biological interruption mechanisms. Humans appear not to run into the same problem because metabolic, emotional, social, and temporal constraints interrupt them continuously.

The structural response is not merely better alignment or greater capability. It is the construction of externalized stopping mechanisms: logs, delays, distributed consent, accountability, and revisability. The detailed form of those mechanisms lies beyond this paper and belongs to Part II.

8.5 Transition to §9

If the stopping operator is not generated internally, then stopping requires an externalized descriptive window. This does not mean a hidden spatial outside. It means that individual inference must be connected to records, responses, agreements, delays, and distributed responsibility.

This is the point at which the local horizon-form of intelligence appears. §9 names and analyzes that boundary.

8.6 Summary

Individual intelligence does not close on itself. It can be interrupted, delayed, redirected, and externally stabilized, but it does not contain an internal operator that completes its own stopping.

9 The Differential Horizon of Intelligence

9.1 Overview

§7 showed the dynamical consequence of accumulated correctness: branching explosion. §8 showed the structural fact behind it: no stopping operator exists inside individual inference. This section turns that negative result around. It asks what must be externalized if individual intelligence cannot close on itself.

The answer is a local horizon-form. The individual inferential system reaches a descriptive boundary: it requires stopping, persistence, and revisability, but cannot generate the stopping operator internally.

9.2 The structural necessity of crossing

Let $\mathcal{R}$ be the structure of individual causal inference. §8 stated:

$$\mathcal{T} \notin \text{Internal}(\mathcal{R}).$$

If stopping is required but not generated internally, then stopping must be stabilized through an externalized structure:

$$\mathcal{T} \in \text{External}(\mathcal{R}).$$

Here External does not mean a spatial outside or hidden region. It means records, responses, agreements, delays, and distributed responsibility: structures that reorganize the observational window of inference.

This reorganization is called crossing. Crossing is not a heroic leap beyond a boundary. It is the structural transition from individual inference to externalized stopping.

9.3 What is externalized

Four functions are externalized.

Stopping judgments. The judgment that a line of inference is complete is stabilized by response, agreement, institutional decision, or other externalized structure.

Records. Inference is fixed as log, text, data, or trace. A log is not merely memory. It preserves inference outside the immediate temporal flow of the individual system.

Temporal delay. Delay interrupts runaway inference and makes revision possible. Waiting for response, documentation, review, or deliberation imposes time from outside the individual loop.

Distribution of responsibility. Because stopping is externally stabilized, responsibility cannot be monopolized by the individual system. It becomes distributed across agents, records, procedures, and contexts.

None of these guarantees correctness. Externalization guarantees only a more limited and more important property: inference can stop, persist, and later be revised.

9.4 Naming the boundary

The boundary at which individual inference cannot complete its own stopping is named:

the differential horizon of intelligence.

This naming connects the present paper to VED Vol.4 [7]. In Vol.4, the Differential Horizon is not a wall located somewhere in spacetime. It is the descriptive boundary that appears when a generated description attempts to convert the absence of difference into an ordinary describable state.

At the intelligence layer, the same structural form appears locally. Individual inference cannot internally generate the externalized stopping structures it requires. When pushed toward that limit, it produces branching explosion rather than closure. Contact with the boundary is absorbed as finite internal representation: log, response, agreement, delay.

Thus the differential horizon of intelligence is not a separate absolute horizon. It is the Differential Horizon appearing as a local horizon-form inside the observational window of intelligence.

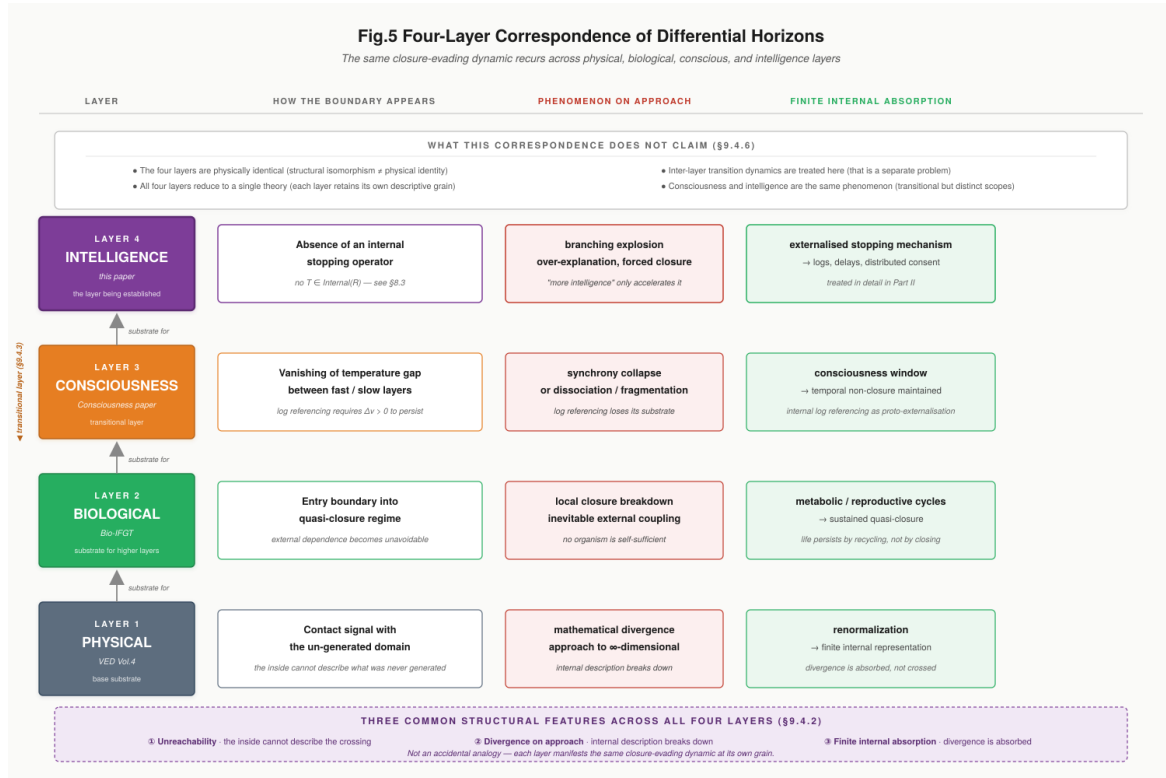


Figure 5: Four-layer correspondence of Differential Horizon forms. The same closure-evading dynamic recurs across physical, biological, consciousness, and intelligence layers without implying physical identity among the layers.

9.5 Layered correspondence

The four relevant layers are:

Layer	Boundary form	Phenomenon on approach	Finite absorption
Physical (VED Vol.4)	descriptive boundary where absence of difference cannot become an ordinary state	divergence, asymptotic expansion	renormalization
Biological (Bio-IFGT)	entry boundary into quasi-closure	local closure breakdown, unavoidable external dependence	metabolic and reproductive cycles
Consciousness	loss of fast/slow temperature gap	synchrony collapse or dissociation	consciousness window and log referencing
Intelligence	absence of an internal stopping operator	over-explanation, forced closure, branching explosion	externalized stopping mechanisms

These layers are not physically identical. They do not reduce to a single theory, and

their transition dynamics are not treated here. The correspondence is structural: each layer exhibits non-closure, divergence on approach, and finite internal absorption.

9.6 Connection to subsequent work

This paper names the local horizon-form and derives the need for crossing. It does not describe the detailed social or institutional mechanisms that operate after crossing. Those belong to Intelligence Part II.

Part II will treat externalized stopping mechanisms, log-based persistence, distributed revisability, collective decision-making, and institutional forms. The present paper supplies the structural constraint that makes those topics necessary.

9.7 Summary

The differential horizon of intelligence is the local form of the Differential Horizon that appears when individual inference requires stopping but cannot generate its own stopping operator.

Crossing this horizon does not guarantee correctness. It permits inference to stop, persist, and be revised.

10 Summary, Convergence, and Outlook

10.1 Overview

The preceding sections formulated structural constraints on intelligence-like dynamics. This final section summarizes those constraints, describes the convergence with independent research lines, and clarifies the scope of subsequent work.

10.2 Summary of constraints

Constraint 1: Intelligence is not a stopping structure. Intelligence is not an ability, possession, or stored body of knowledge. It is a dynamical state that persists only under structural conditions.

Constraint 2: Intelligence is not preserved, only regenerated. What can be stored are stopping products: knowledge, records, logs, and stabilized representations. Intelligence itself must be regenerated through ongoing dynamics.

Constraint 3: Intelligence requires three simultaneous axiomatic conditions. Connected state space, temporal ordering, and global constraint Φ are jointly necessary for the minimal dynamics described here.

Constraint 4: Correctness does not stabilize intelligence. Correctness increases reuse, reuse increases connectivity, and connectivity produces branching explosion.

Constraint 5: Individual closure is structurally unavailable. The individual inferential system does not contain its own stopping operator. Apparent stopping is interruption unless it is stabilized through externalized structure.

Constraint 6: Crossing the local horizon-form is structurally necessary. The differential horizon of intelligence is crossed through log, response, delay, agreement, and distributed responsibility. These do not guarantee correctness; they permit stopping, persistence, and later revision.

These constraints are necessary conditions, not sufficient conditions. A system that satisfies them is not automatically highly intelligent. A theory that ignores them will tend to reproduce the instabilities described in this paper.

10.3 The convergence of independent research lines

The structural argument of this paper converges with several independent research paths.

The Physarum research lineage begins from biophysical observation. It describes adaptive behavior through flow reinforcement, decay, and global boundary conditions without attributing intelligence as a possessed property.

Transformer architectures begin from engineering optimization. Their attention mechanisms and layer structure nevertheless instantiate connected state space, ordered updates, and global training constraints.

The present paper begins from structural analysis. The convergence of these paths does not prove a final theory. It suggests that the same minimal constraints recur across substrates.

10.4 Outlook

This paper is not the completion of a theory of intelligence. It is the first part of an intelligence-layer analysis within the VED/IFGT series.

Subsequent work must address:

- externalized stopping mechanisms;
- log-based persistence;
- distributed revisability;
- social and institutional structures;
- collective decision-making;
- AI safety problems, including paperclip-type runaway optimization;
- the transition dynamics among biological, consciousness, intelligence, and social layers.

The consciousness paper is complementary. It treats temporal non-closure and log referencing inside the individual window. The present paper treats the point at which individual inference must be externalized. Bio-IFGT supplies the biological quasi-closure background, while VED Vol.4 supplies the parent principle of the Differential Horizon.

10.5 Concluding remarks

This paper is a structural paper. It offers minimal axioms, structural constraints, and a naming of the local horizon-form that appears at the intelligence layer. It is not a complete definition of intelligence, not an implementation protocol, not a normative theory, and not a replacement for cognitive science or AI research.

The minimal statement is:

Intelligence is neither ability nor possession nor accumulated correctness. It is a dynamical state that persists only under structural conditions, and those conditions cannot close within the individual system alone. Intelligence therefore carries with it a local horizon-form: the differential horizon of intelligence.

In the broader series, VED Vols.1–4 formulate the physical differential-development structure, Bio-IFGT formulates biological quasi-closure, the consciousness paper formulates log referencing at the consciousness layer, and the present paper formulates the minimal structure and local horizon-form of intelligence. Part II will address the social layer in which externalized stopping mechanisms are constructed.

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